INCH-POUND
A-A-59767A
October 14, 2022
SUPERSEDING
A-A-59767
February 3, 2005

COMMERCIAL ITEM DESCRIPTION

LIFE PRESERVER, MK-1

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers requirements for the MK-1 life preserver for use on U.S. Navy surface ships and submarines by personnel performing flight deck and any other shipboard operations (except for operations that could damage the inflatable bladder, e.g., hot work).

2. CLASSIFICATION. The MK-1 life preserver vest cover shall conform to the following sizes and colors.

2.1 Sizes. The MK-1 life preserver vest cover shall be available in small, medium, large, X-large, and 2X-large sizes (see table I). Unless otherwise specified herein, all measurement tolerances shall be ±¼ inch.

TABLE I. Sizes and measurements.

MEASUREMENTS			
Size Label		**Chest (inches) (Below Armhole)**	**Back Length (inches) (Center-Back)**
Small	36 to 38	42 – 43	24 – 25
Medium	40 to 42	46 – 47	24 – 25
Large	44 to 46	50 – 51	24 – 25
X-Large	48 to 50	54 – 55	25 – 26
2X-Large	52 to 54	58 – 59	25.5 – 26.5

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 4220

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2.2 Colors. The MK-1 life preserver vest cover shall be available in 7 colors: white, red, green, blue, brown, yellow, and purple.

3. SALIENT CHARACTERISTICS.

3.1 Physical and performance characteristics. In man overboard emergencies, the MK-1 life preserver shall, upon water entry, automatically inflate, self-right the wearer, and provide good torso orientation and head support, even if the wearer is unconscious. The MK-1 life preserver shall be comprised of a flame-resistant vest cover (see 3.1.2) and an inflatable bladder (see 3.2.2).

3.1.1 Flotation and buoyancy. The MK-1 life preserver shall comply with the flotation and buoyancy, quality assurance requirements for a type III personal flotation device (PFD) as specified in 46 CFR § 160.076 and UL 1180.

3.1.2 Fire retardance and flame resistance. The vest cover material shall be fire-retardant treated, 88 percent cotton and 12 percent nylon, and in a twill-weave in the colors listed in 2.2. The material flame resistance properties in terms of char length shall not exceed 5 inches and have a maximum after-flame of 2 seconds when tested in accordance with ASTM D6413/D6413M.

3.1.3 Breaking strength. The vest cover material shall be a minimum of 9 ounces per square yard when tested in accordance with ASTM D3776/D3776M; additionally, the vest cover material shall have a minimum breaking strength of 130 pounds in the warp and 100 pounds in the filling when tested in accordance with ASTM D5034.

3.1.4 Tear strength. The vest cover material shall have a minimum tear strength of 7 pounds in the warp and 6½ pounds in the filling when tested in accordance with ASTM D1424.

3.1.5 Cloth shrinkage. Cloth shrinkage of the vest cover material shall not exceed 3 percent in each direction when tested in accordance with AATCC 96.

3.2 Design and construction.

3.2.1 Vest cover. The vest cover shall be cardigan style, sleeveless, with a front zipper closure (heavy duty plastic), and a closure (hook-and-loop fastener or snap-and-zipper keeper) at the top of the zipper. The inside vest cover shall be two-ply in the front and upper back to accommodate the bladder. The upper back shall contain foam cushioning. An adjustable waist strap shall completely encircle the vest cover and be secured in the front with a slide buckle fastener. The right front of the vest cover shall have two openings, one sized for the oral inflation tube and the other for the bladder manifold with attached inflator and 24-gram CO_2 cylinder. A securing strap shall be provided for the cylinder. The upper-right front of the vest cover shall have a flap secured by hook-and-loop fastener at the top and side to cover the oral inflation tube. The lower-right front of the vest cover shall have a foam-cushioned flap to cover the inflator and 24-gram CO_2 cylinder. The face of the flap shall contain the printing "WARNING! AUTOMATIC INFLATION ASSEMBLY INSTALLED. DO NOT USE ABOARD AIRCRAFT." The vest cover design shall provide channels for routing the man overboard indicator (MOBI) antenna internally up the vest cover collar without affecting the transmission signal. The back of the vest cover shall contain a 1-inch high by 3½-inch (-0/+½ inch) wide opening to allow a safety harness D-ring to fit through the preserver. See figures 1, 2, and 3 for details on the vest cover.

3.2.2 Inflatable bladder. The inflatable bladder contained within the MK-1 life preserver shall meet type III PFD performance requirements as specified in 46 CFR § 160.076 and UL 1180, which includes all physical material components, construction, testing, quality assurance, and flotation and buoyancy characteristics. The right side of the inflatable bladder shall contain an oral inflation tube and manifold for the inflator device that is compatible with the vest cover. The CO_2 cylinder size (24 grams), model, and National Stock Number (NSN) shall be printed on the inflatable bladder. The inflatable bladder shall be internally secured to the vest cover. See figure 4 for details on the inflatable bladder.

3.2.2.1 Inflatable bladder securing straps. The vest cover, internally (left and right front), shall have minimum ¾-inch wide nylon, polyester, or polypropylene straps with stud-and-eyelet snap attachments to attach the inflatable bladder.

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3.2.3 Automatic inflator and cylinder. The inflatable bladder shall contain a water-activated automatic inflator mechanism (see 7.6) and a 24-gram CO_2 cylinder. A label detailing the operation of the inflator and cylinder shall be attached inside the inflator cover.

3.2.4 Waist belt. The waist belt shall be minimum 1-inch wide nylon, polyester, or polypropylene webbing with a slide adjuster and secured with a slide buckle closure.

3.2.5 Reflective tape. The vest cover shall contain launderable reflective tape on the left and right fronts and across the upper back. The reflective tape shall be U.S. Coast Guard approved Safety of Life at Sea (SOLAS) retro-reflective material. The tape shall be minimum 2 inches wide by 4½ inches long on each chest and minimum 2 inches wide by 12 inches long across the back.

3.2.6 Distress marker light patch. The upper-left section (shoulder area) of the vest cover shall contain a minimum 1½- by 1½-inch nylon loop tape patch to temporarily attach the distress marker light in man overboard situations. The distress marker light shall have a ¾- by 1¾-inch nylon hook tape patch adhered to the back of the light.

3.2.7 Accessories pockets. The vest shall have two double-ply pockets sized to hold the distress marker light, flat whistle, sea dye marker, and MOBI (see 7.5). One pocket (with two chambers) shall be located on the upper-left front of the vest and contain the whistle and the distress marker light. The other pocket (with two chambers) shall be located on the lower-left side and contain the sea dye marker and the MOBI (see 7.5). The distress marker light pocket and whistle pocket shall have two eyelets near the top of the pocket for cord attachment to the distress marker light and whistle. The sea dye marker and MOBI pocket shall have an eyelet near the top for the sea dye marker; it shall also have a hole in the top to allow the passage of the MOBI antennae and one in the side to allow the MOBI to be turned on or off. The pockets shall have flaps secured by hook and loop.

3.2.8 Accessories securing cords. The distress marker light, sea dye marker, and flat whistle shall be secured to the accessories pocket with a nominal 1/16-inch diameter nylon, polyester, or polypropylene cord, having a minimum tensile strength of 90 pounds. The cord shall be 24 inches in length for the flat whistle and 48 inches in length for the distress marker light and sea dye marker. Cord length tolerances shall be ±2 inches.

3.2.9 Labeling. The inside of the vest shall contain labeling indicating the manufacturer name, vest model number, National Stock Number (NSN), and size.

3.3 Compatibility. The complete vest (outer cover, bladder, and accessories components) shall be totally interchangeable with the components of the Stearns Inc., Model I441 MK-1 life preserver; the Mustang Survival, Model MD3040 life preserver; and the Switlik/Eastern, Model CMV 1001 life preserver.

3.4 Environmental requirements. All components, with the exception of the flame-resistant vest cover, shall be Underwriters Laboratories (UL) recognized for use in PFDs and shall meet the requirements for accelerated weathering in accordance with UL 1191.

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. PRODUCT CONFORMANCE PROVISIONS. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

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7. NOTES.

7.1 Source of documents.

7.1.1 American Association of Textile Chemists and Colorists (AATCC) Standards. AATCC standards are available from www.aatcc.org or the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.

7.1.2 ASTM Standards. ASTM standards are available from www.astm.org or ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959.

7.1.3 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov or the Superintendent of Documents, U.S. Government Publishing Office, Washington, DC 20401.

7.1.4 UL Standards. UL standards are available from www.ul.com or COMM 2000, 151 Eastern Avenue, Bensenville, IL 60106.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. NSNs.
c. Product conformance provisions.
d. Quantity required.
e. Packaging requirements.

7.3 National stock numbers (NSN). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	ITEM	NSN	ITEM
4220-01-487-2926	Life Preserver Bladder	4220-01-487-2940	MK-1 Life Preserver - Small White
4220-01-470-9906	Inflator Mechanism	4220-01-487-2950	MK-1 Life Preserver - Small Red
4220-01-487-2878	24-Gram CO_2 Cylinder	4220-01-487-2963	MK-1 Life Preserver - Small Green
6230-01-686-4184	Distress Marker Light	4220-01-487-2972	MK-1 Life Preserver - Small Blue
4220-01-487-2931	Whistle	4220-01-487-2978	MK-1 Life Preserver - Small Brown
6850-00-270-9986	Sea Dye Marker	4220-01-487-2980	MK-1 Life Preserver - Small Yellow
4220-01-487-3192	MK-1 Life Preserver - Medium White	4220-01-487-3300	MK-1 Life Preserver - Large White
4220-01-487-3201	MK-1 Life Preserver - Medium Red	4220-01-487-3619	MK-1 Life Preserver - Large Red
4220-01-487-3206	MK-1 Life Preserver - Medium Green	4220-01-487-3625	MK-1 Life Preserver - Large Green
4220-01-487-3210	MK-1 Life Preserver - Medium Blue	4220-01-487-3647	MK-1 Life Preserver - Large Blue
4220-01-487-3239	MK-1 Life Preserver - Medium Brown	4220-01-487-3650	MK-1 Life Preserver - Large Brown
4220-01-487-3294	MK-1 Life Preserver - Medium Yellow	4220-01-487-3652	MK-1 Life Preserver - Large Yellow
4220-01-487-3297	MK-1 Life Preserver - Medium Purple	4220-01-487-3655	MK-1 Life Preserver - Large Purple

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NSN	ITEM	NSN	ITEM
4220-01-487-3657	MK-1 Life Preserver - X-Large White	4220-01-504-8263	MK-1 Life Preserver - 2X-Large Yellow
4220-01-487-3659	MK-1 Life Preserver - X-Large Red	4220-01-504-8268	MK-1 Life Preserver - 2X-Large Red
4220-01-487-3661	MK-1 Life Preserver - X-Large Green	4220-01-504-8270	MK-1 Life Preserver - 2X-Large White
4220-01-487-3663	MK-1 Life Preserver - X-Large Blue	4220-01-504-8272	MK-1 Life Preserver - 2X-Large Brown
4220-01-487-3666	MK-1 Life Preserver - X-Large Brown	4220-01-504-8277	MK-1 Life Preserver - 2X-Large Purple
4220-01-487-3667	MK-1 Life Preserver - X-Large Yellow	4220-01-504-8278	MK-1 Life Preserver - 2X-Large Blue
4220-01-487-3669	MK-1 Life Preserver - X-Large Purple	4220-01-504-8280	MK-1 Life Preserver - 2X-Large Green

7.4 Key words.

Buoyancy

Flame-Resistant

Flight Deck Life Vest

Flotation

Self-Righting

7.5 Man overboard indicator (MOBI). MOBI is not included with the MK-1 life preserver and is purchased separately.

7.6 Automatic inflator mechanism. Automatic inflator mechanism is not included with the MK-1 life preserver and is purchased separately.

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FIGURE 1. Front view of MK-1.



FIGURE 2. Back view of MK-1with D-ring hole.

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FIGURE 3. View of MK-1 accessories.

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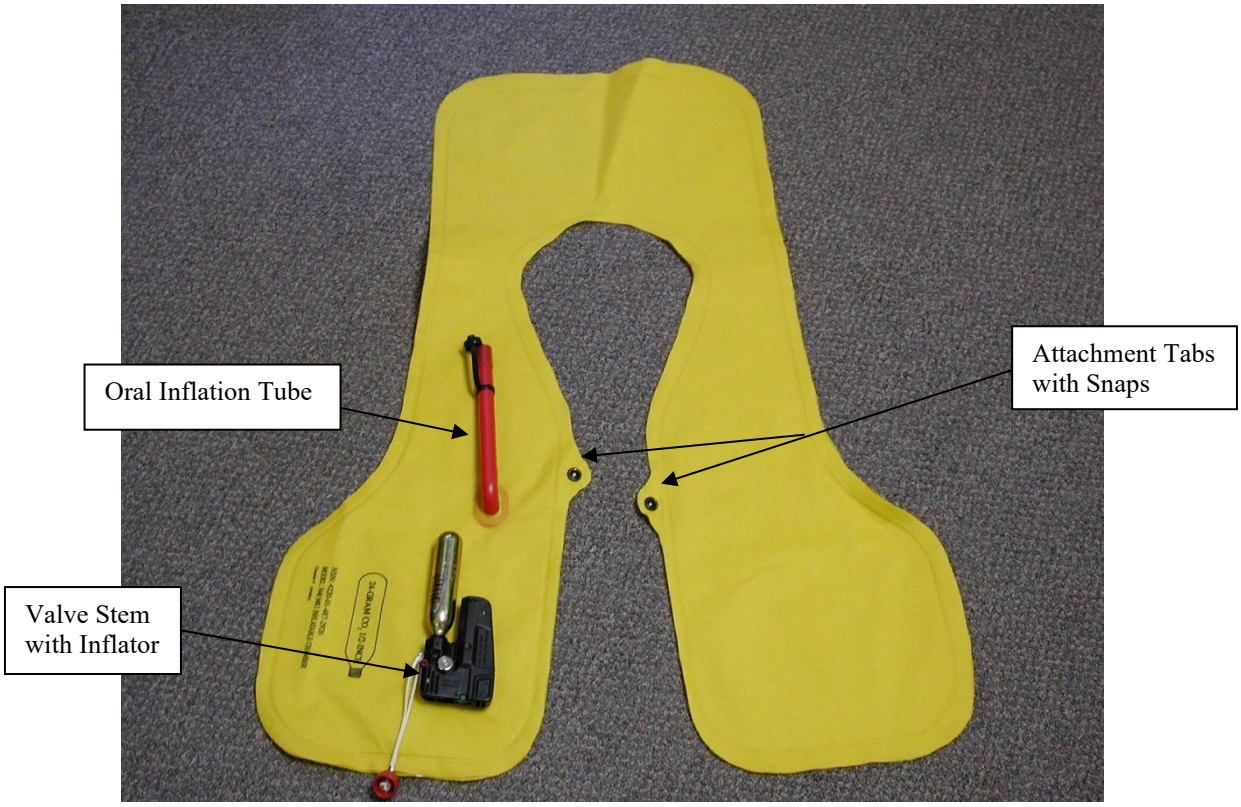


FIGURE 4. View of MK-1 inflatable bladder.

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Navy – SH
Air Force – 03

Preparing Activity:
Navy – SH
(Project 4220-2022-002)

Review Activity:
Navy – AS

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